Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and pre sent potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclo sed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI-Based Crop Cultivation Guidance Solution Using Satellite Imagery Analysis**			
Korea's 12 major national strategic technology fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (Major Category)**		**Technology Field 2 (Major Category)**	
	Information & Communication		Information & Communication	
	Medium Category	**Medium Category**	**Small Category**	**Medium Category**
	Service Platforms and Application Software	Software	Software Solutions	Service Platforms and Application Software
Proposed technology concept	**Core Concept** The proposed technology is a real-time, satellite-based agricultural decision-support platform that leverages artificial intelligence to analyze multispectral imagery collected from a large-scale satellite network. By integrating data from 220 satellites across 12 satellite types, the system generates continuous, high-accuracy assessments of crop conditions throughout the entire cultivation cycle—from planting to harvest. The core concept is to deliver a scalable, data-driven cultivation intelligence service that transforms complex satellite imagery into actionable insights for farmers, enabling optimized resource use, reduced operational uncertainty, and measurable yield improvement at a globally competitive cost.			

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	Key Functions and Potential Applications ● Key Functions ■ Crop Classification and Mapping · Automatically distinguishes crop types across large agricultural areas using multispectral and temporal image data, enabling precise farm-level and regional crop monitoring. ■ Field-Level **Monitoring with Daily Updates** · Provides real-time tracking of soil moisture, soil nitrogen, chlorophyll levels, and eight other key agricultural indices with up to 98% accuracy, enabling early detection of stress, nutrient deficiencies, and environmental risks. ■ **AI-Based Smart Cultivation Management** · Generates customized AI-driven cultivation guides optimized for each field, supporting decision-making on irrigation, fertilization, pest control, and harvest timing. ■ **Multi-Satellite Data Fusion** · Integrates imagery from 220 satellites to deliver robust, cloud-resilient monitoring unaffected by local weather or sensor limitations. ● Potential Applications ■ **Farm-Level Productivity Enhancement** · Achieves measurable improvement in crop yield (average +18%) by providing continuous insights and precise management recommendations. ■ **Regional and National Agricultural Planning** · Supports policy-making, food security programs, and resource allocation by offering high-resolution, real-time crop distribution and growth status data. ■ **Sustainable Agriculture and ESG Monitoring** · Enables monitoring of soil health, water use efficiency, land degradation, and environmental impact for sustainable farming programs and carbon credit initiatives. ■ **Global Scalability for Developing and Emerging Markets** · Applicability in over farms in Korea and across international regions, demonstrating suitability for diverse climates and agricultural systems.
Rationale for Joint R&D Project	**Critical Technical Barriers** Large-scale open-field farming regions face common challenges: labor shortages, difficulty in uniform crop management across vast areas, and unpredictable yields due to climate variability. While it is proven to be effective in Asian markets, expansion into North American and other advanced agricultural markets requires localized AI model development for region-specific crops and climate conditions. Global joint R&D with partners will enable calibration of satellite imagery analysis algorithms for local agricultural environments, validation of agricultural indices under different climate conditions, and development of crop-specific AI models for crops including corn, wheat and soybean.

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	Why Global Collaboration? While existing solutions like Climate FieldView (Bayer) and Farmers Edge offer satellite-based monitoring, they primarily target large enterprise farms with premium pricing. The proposed technology differentiates through: (1) competitive cost structure ($12/ha), enabling access for mid-sized farms; (2) fully web-based service requiring no additional hardware installation or software setup that farmers can access all features through any web browser without purchasing sensors, drones, and specialized equipment.
Key words	1. Satellite 2. Precision Agriculture 3. Agricultural AI Analytics 4. Yield Prediction

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (5)	End of the Project	Level (7)

R&D Objectives

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This project aims to develop localized AI-powered crop monitoring and yield prediction technology for the US Midwest agricultural market.

Detailed Objectives

1. Region-Calibrated Agricultural Index System Adapt and validate 9 agricultural indices (NDVI, NDWI, Soil Moisture, Nitrogen Level, Chlorophyll Index, LAI, Canopy Temperature, GNDVI, EVI) for US Midwest climate and soil conditions. This requires recalibration of index thresholds and algorithms based on local ground-truth data.
2. Crop-Specific AI Yield Prediction Models Develop machine learning models tailored to three major US crops:
   - Soybean: Focus on water stress detection and Iron Deficiency Chlorosis (IDC) identification during R1-R6 growth stages
   - Corn: Emphasize nitrogen stress monitoring and heat stress detection during critical VT-R1 pollination period
   - Wheat: Enable winter survival assessment and disease (rust) early detection during heading stage
3. US-Localized Farm Monitoring Platform Build a web-based dashboard providing real-time field visualization, automated alert systems, and bi-weekly yield prediction reports optimized for US farming practices and regulatory requirements.

R&D Target Specifications

Specification	Target Value	Measurement Method
Yield Prediction Accuracy	≥85%	Comparison with actual harvest data
Crop Classification Accuracy	≥95%	Validation against USDA crop layer data
Agricultural Index Validation	100% completion for 3 crops	Field measurement correlation (R^2 ≥0.8)
Daily Monitoring Coverage	25,000+ acres	Platform data logging
Alert Response Time	Within 24 hours of anomaly detection	System timestamp verification
AI Model Development	3 crop types (Soybean, Corn, Wheat)	Model deployment confirmation
Project Duration	12 months	-

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R&D Plan

Phase	Tasks	Milestone	Timeline
Business Analysis & Planning	Define project scope, requirements analysis, crop calendar mapping for Soybean/Corn/Wheat, baseline data collection (yield, soil, weather), system architecture design	Functional specification and baseline dataset completed	Month 1-3
Satellite Matching & Index Calibration	Select optimal satellite combinations per crop, calibrate index thresholds using ground-truth data, establish automated imagery processing pipeline	All 9 agricultural indices validated ($R^2 \geq 0.8$)	Month 3-6
AI Model Development	Train crop-specific ML models: Soybean (water stress/IDC), Corn (nitrogen/heat stress), Wheat (winter survival/disease); develop US-localized web monitoring platform	3 AI models deployed; platform beta released	Month 5-9
Validation & Optimization	Collect actual harvest data, assess prediction accuracy, analyze error patterns, refine models	Yield prediction accuracy ≥85% achieved; final report delivered	Month 9-12

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute

Area	Responsibilities
Field Access	Provide access to corn and wheat fields (10,000+ acres) in the US Midwest region
Data Provision	Supply ground-truth data including: soil samples, irrigation records, fertilization history, actual yield records, and local weather station data
Validation	Validate satellite-derived agricultural indices against field measurements; provide periodic feedback (monthly) for AI model calibration
Market Support	Facilitate connections with local farmers and provide insights on US agricultural market requirements

R&R of Korean SME partners

Area	Responsibilities
Satellite Platform	Provide access to 220 satellites (12 types); deliver daily multispectral imagery and 9 key agricultural indices (soil moisture, nitrogen levels, chlorophyll index, etc.)
AI Model Development	Develop crop-specific AI models for corn and wheat; achieve ≥85% yield prediction accuracy
Web Monitoring System	Develop and operate US-localized web monitoring dashboard; provide real-time field status visualization and alert notifications
API Integration	Build and maintain API infrastructure for seamless data exchange; enable integration with partner's existing farm management systems
Reporting	Deliver periodic yield prediction reports (bi-weekly during growing season); provide end-of-season harvest forecast and anomaly analysis reports

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Technical Impact: ■ Achieve high prediction accuracy through fusion of multi-satellite data and AI technology ■ Establish immediate response system based on daily updated data for rapid weather event response ■ Enable zone-specific precision management through detailed field monitoring ● Economic Impact: ■ Minimize damage from seasonal drought and flooding through differentiated irrigation and fertilization ■ Reduce water and fertilizer usage through precision application ■ Expected average 18% yield increase through individual farm monitoring and agricultural decision-making tools ■ Risk reduction through advance preparation for yield decrease caused by weather anomalies ● Scalability ■ Crop monitoring and yield prediction models developed in pilot region can be extended to other crops in the same region or same crops in other US regions ■ The pilot project targets 25,000 acres; the total addressable market in the US Midwest exceeds 127 million acres of agricultural land (USDA Climate Hubs), representing significant scale-up opportunity ■ Long-term global expansion possible to countries with similar climates: Canada, Australia, Argentina ● Market Potential ■ Global precision agriculture market growing rapidly due to labor shortages and climate change ■ Low cost structure ($12/ha) enables broad market penetration

6. References

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